

Synthetic Biology I
FS26



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FIFTY SHADES OF STRESS

Bilal, 22

His typhoid infection used to be cured with simple antibiotics.

Image taken after his stay in a hospital, taking a very strong and expensive antibiotic to survive.

Antibiotic Resistance

→ Bacteria no longer respond well to medicines, making infections harder to treat and increasing the risk of severe illness and death.

Direct cause of **1.27 million deaths** (2.3% of all deaths) in 2019

Additional health care costs of \$66 billion by 2024, \$159 billion by 2050

Does not care about borders

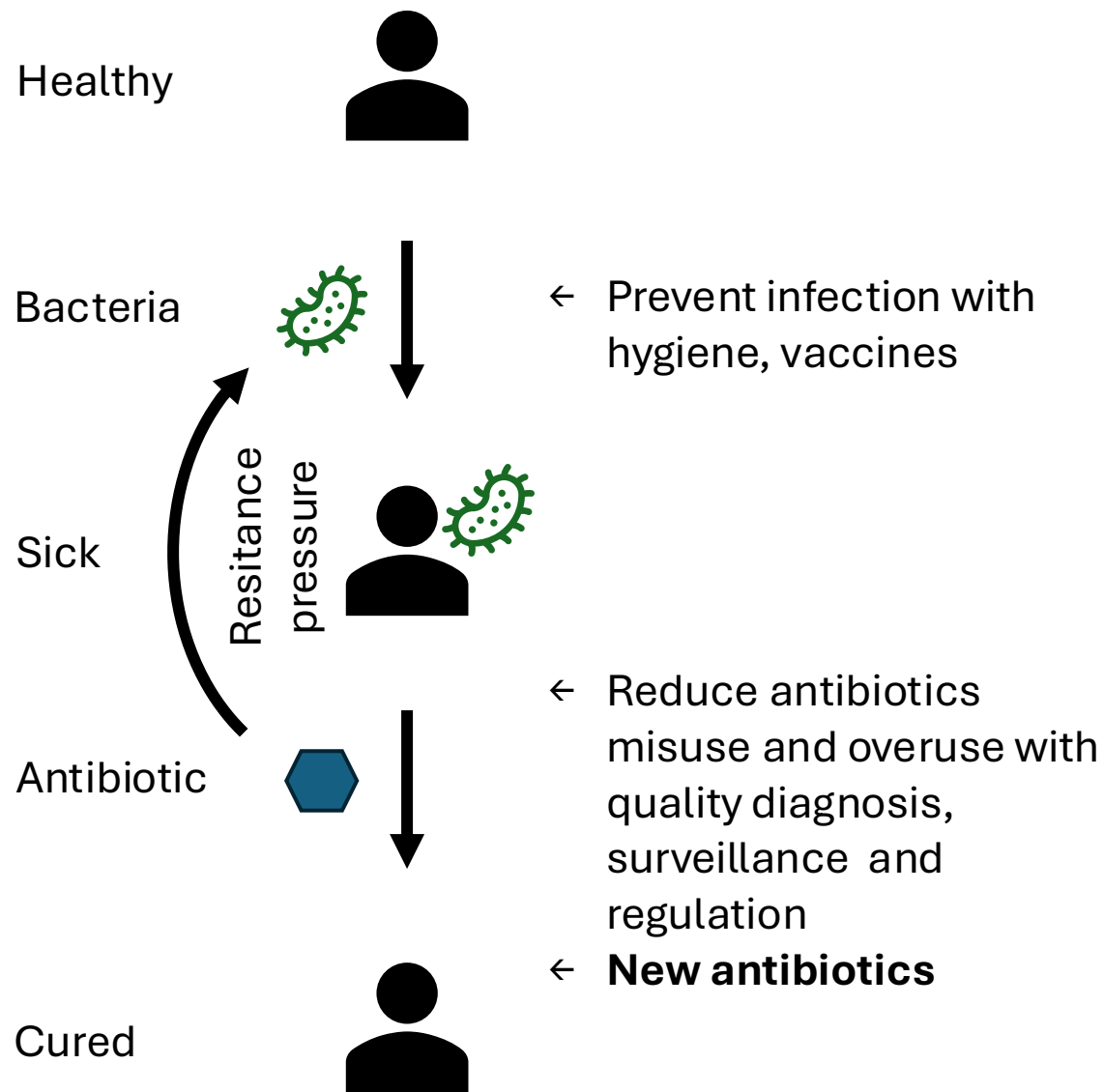
Threatens:

Treating common infections

Ability to do surgery / chemotherapy



Solving Antibiotic Resistance



[6], [7]

We are Losing the Armsrace:

- New antibiotics R&D pipeline crisis

New drugs might not overcome existing resistance mechanisms [6]

Over the past 6 years, only 16 new antibacterials have received market approval [6]

Fragile antibacterial pipeline [6]

Market not incentivised

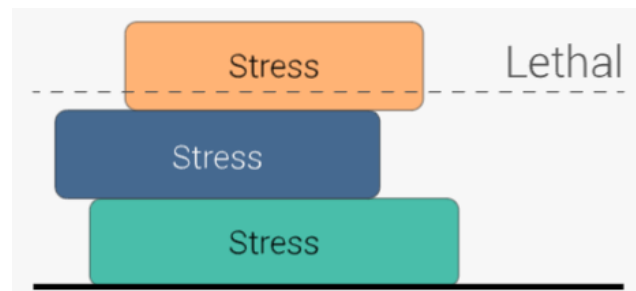
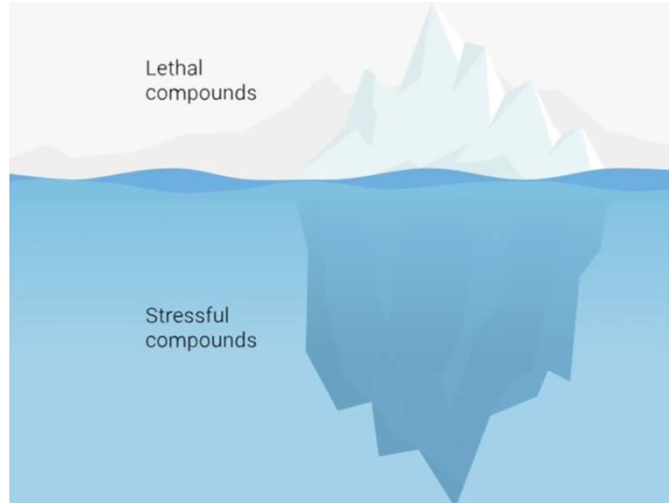
- Increasing resistance

from 2018 to 2023, resistance increased in over 40% in monitored antibiotic treatments [5]

In 2023 one in six bacterial infections worldwide were caused by bacteria resistant to antibiotics [5]

How Does the Team Plan
to Solve the Problem?

Core Idea

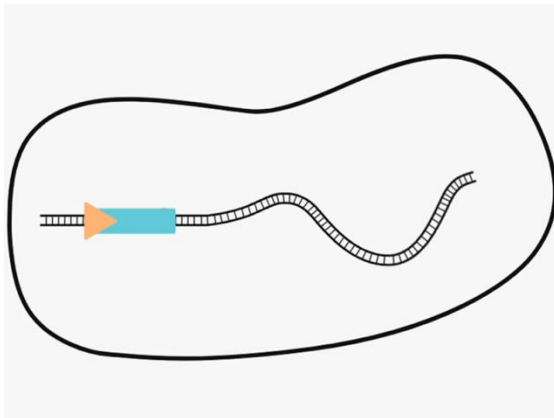
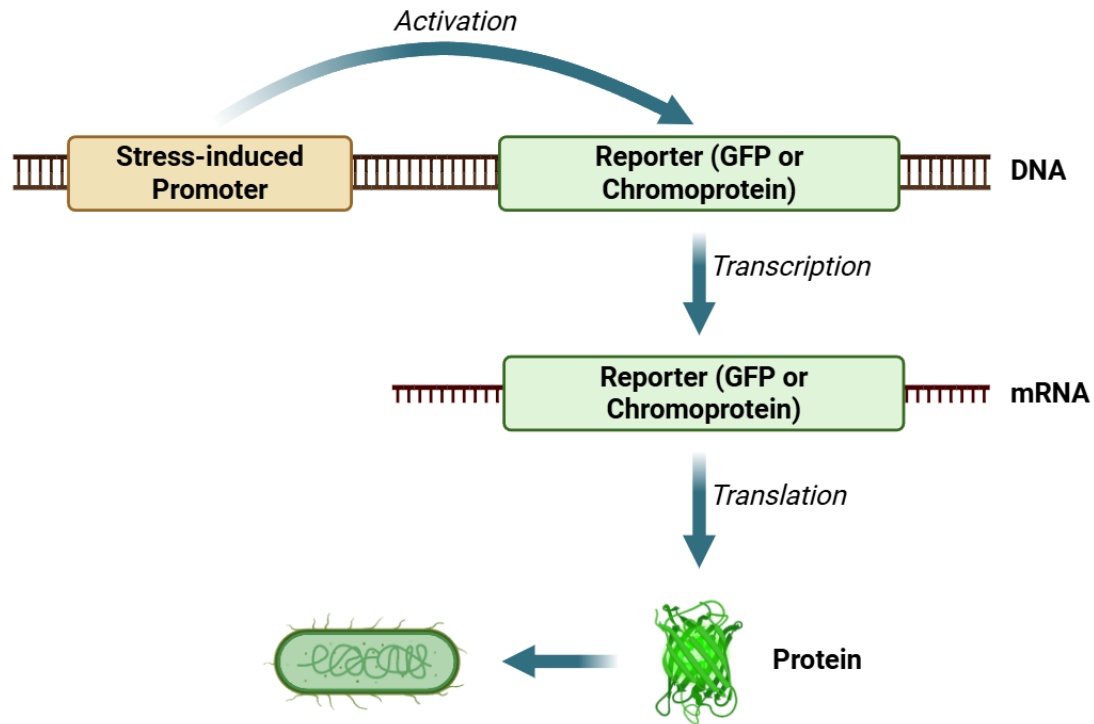


50 S.O.S aims to solve those Problems by:

- Identification of **microbial stressors** rather than just lethal compounds.
- **Combination** of stressors for lethal effect
- Identify the **mode of action** early to create effective combinations

→ Decrease resistance formation by combination therapy

Design Approach



- Coupling unique **visual markers** to stress activated promoters
- Stress promoter initiates molecular response to external stress
- Compound induces stress → the promotor gets activated (**input sensor**) → produces the visual reporter protein (**output signal**)
- Reporter proteins:
 - Fluorophore → emits specific wavelength
 - Chromoprotein → produces color
- **Unique reporters** can be used for different stress responses to elucidate **mode of action**

Design Approach

Advantages

- Follows standard genetic circuit architecture:

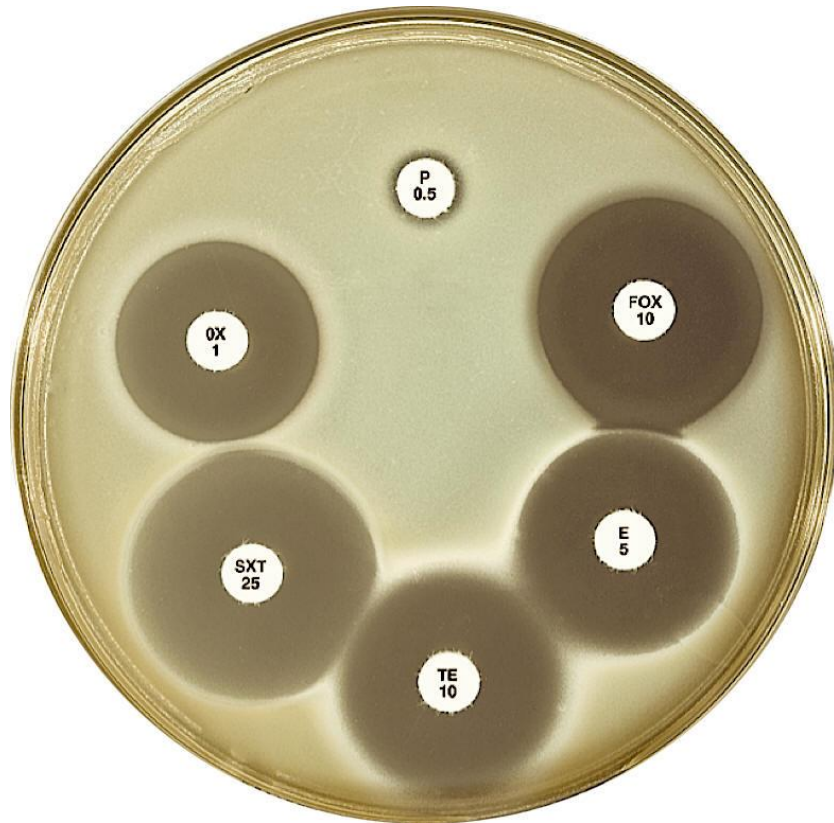
Input → System → **output**

- Detection of **stressors** & information on **mode of action** → combination therapy
- Compatible with **library-based screening**
- **Visual output** → reduces need for complex instrumentation
- Modular design

Potential Problems

- **No feedback control** → completely open loop
- **Promoter specificity** might be incomplete or overlapping

Traditional Testing



New compounds are usually tested in a **Disk Diffusion Test**:

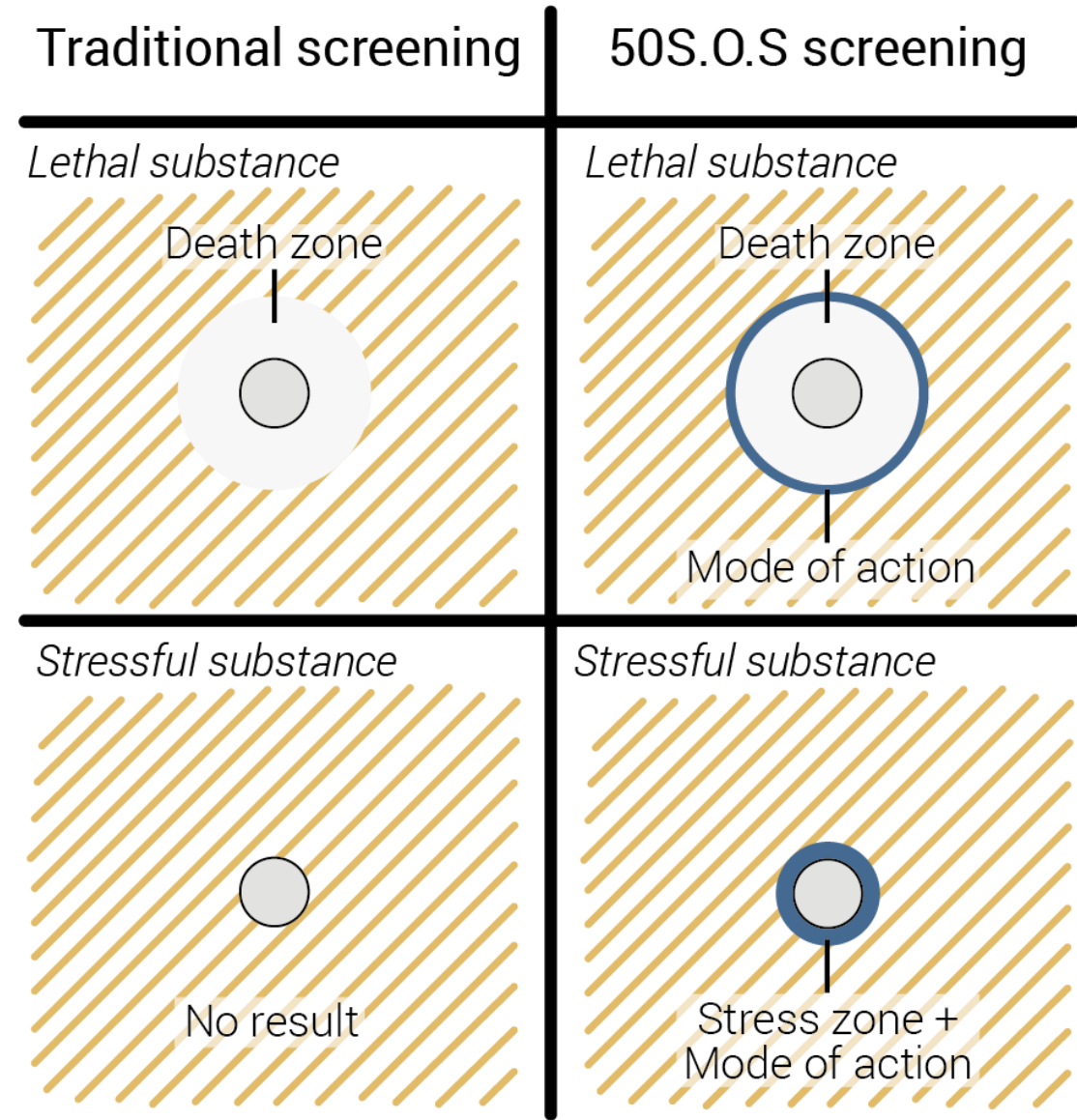
1. Compound is placed on a “lawn” of target bacteria
2. Compound **diffuses** away from its source into the agar creating a **concentration gradient**
3. If the compound is antibiotic, it will inhibit the bacteria from growing in a circle surrounding its source
4. The extend of this **Inhibition Zone** is dependent on the **sensitivity** of the bacteria to the compound

Experimental Setup

Improved compound screening by using *E. coli* carrying the engineered promoter-reporter constructs:

1. For **antibiotic substances**, the edge of the “death zone” indicates mode of action
→ Non-lethal concentrations

2. **Stressful** substances can now be detected including their mode of action



How did they Model their
System?

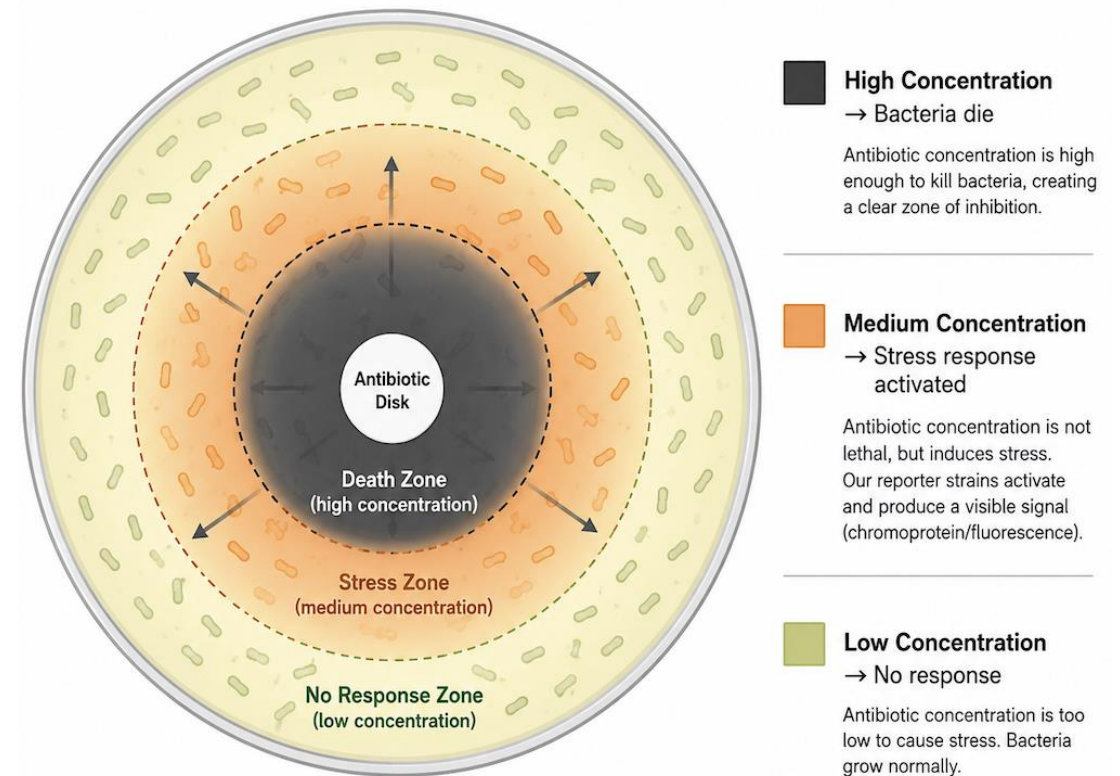
Modelling Antibiotic Diffusion to Predict Bacterial Stress

Problem:

- Antibiotics spread through agar over time
- Stress reporter bacteria respond differently for varying concentration

Need to predict:

- Where bacteria experience stress
- When stress becomes visible
- How to optimize disk placement

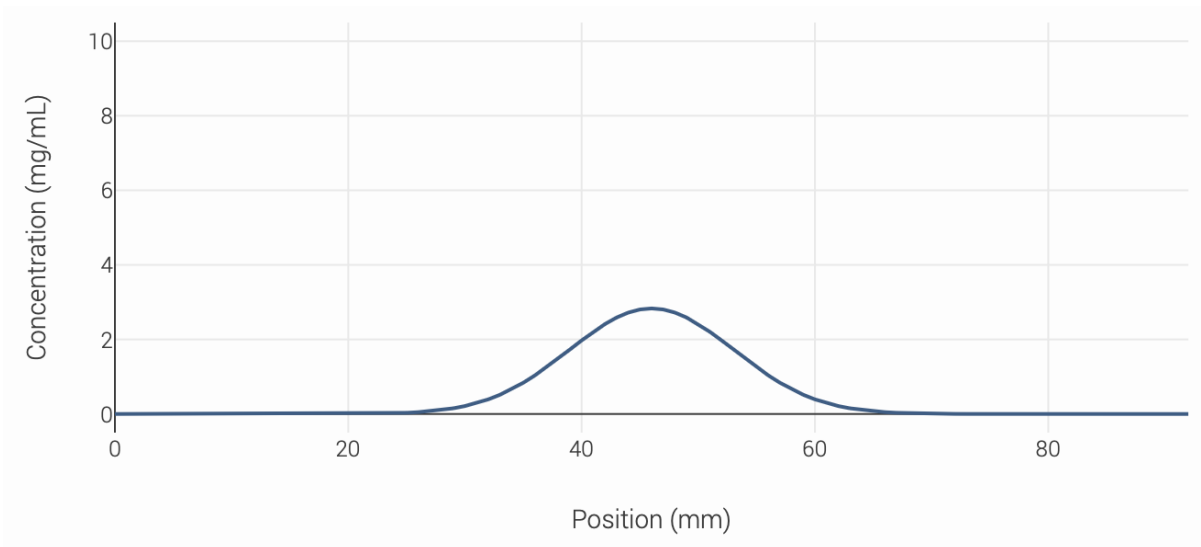


Mathematical Model

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial x^2}$$

c = Antibiotic concentration

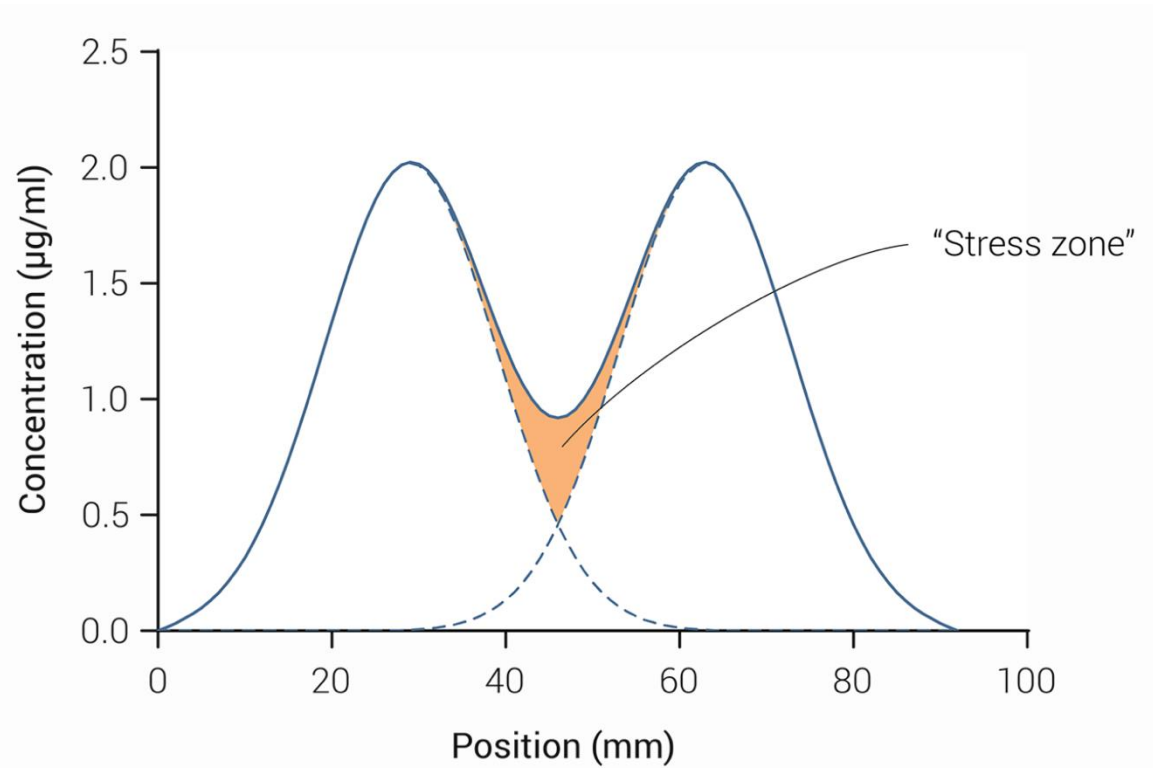
D = Diffusion coefficient



[8], [A]

- Used Fick's 2nd Law to model diffusion of antibiotic in agar
- Minimum inhibitory concentration & diffusion coefficient D estimated with experiments
- Differential equation solved with **finite differences approach** to discretize space

Creating “Stress Zones”

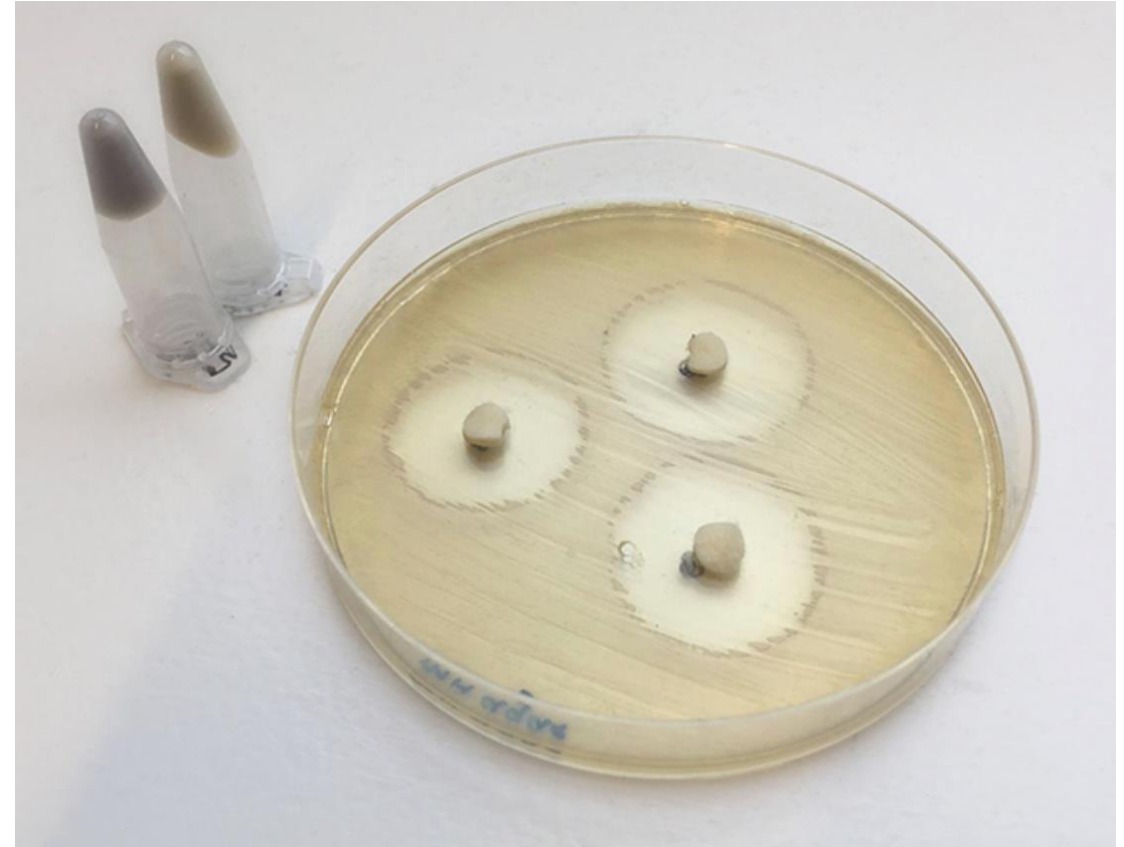


- **Goal:** Create a larger region where bacteria are stressed but alive
- One antibiotic disk only creates thin stress ring around inhibition zone
- **Hypothesis:** Overlapping diffusion gradient from multiple disks could enlarge region

- Model predicted overlapping diffusion gradients would create **larger “stress zones”**
- Experimentally, no strong central stress zone observed

Possible reasons:

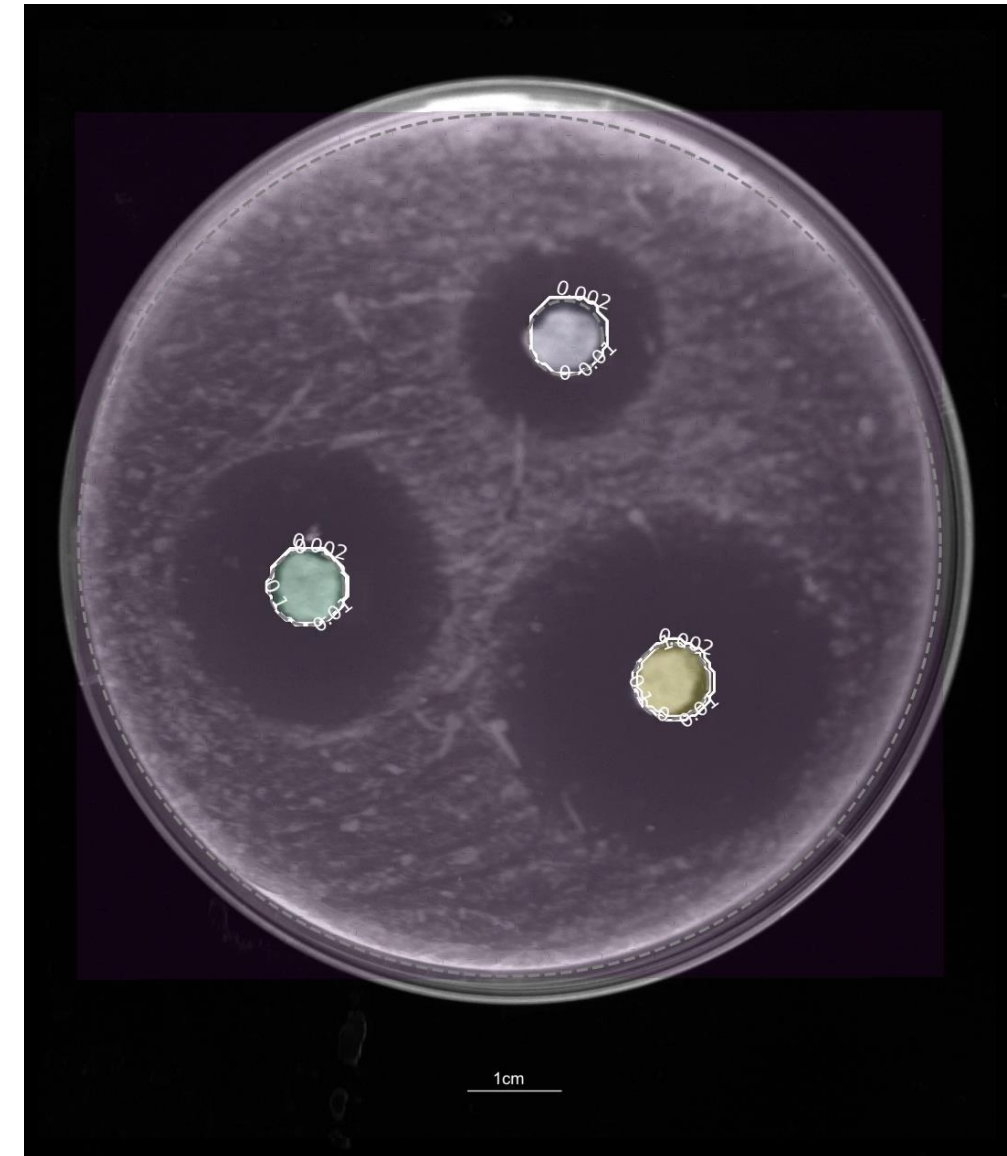
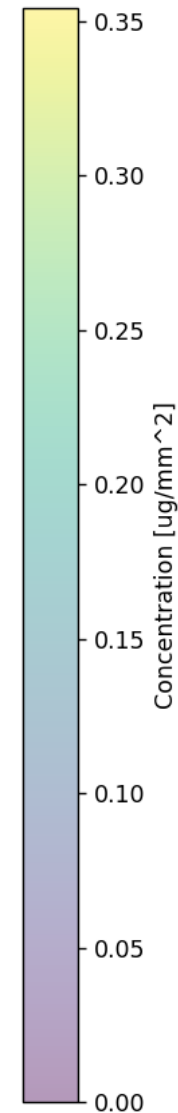
- Stress window likely very narrow
- Stressed cells grow slowly & produce less protein
- Chromoproteins (AmilCP) mature slowly



Our own Experiment + Simulation

- Bacterial lawn (*E. coli* DH10 β) on 1% agar
 - Add antibiotic (ampicillin) discs with $2\mu\text{g}$, $10\mu\text{g}$, $50\mu\text{g}$
 - Wait 17.5h and image resulting inhibition rings
 - Simulate to validate their model.
- **Simulation:**
 - $\frac{\partial c}{\partial t} = D\nabla^2 c$ Homogeneous 2D diffusion equation
 - Discretization by finite differences
 - IC: various concentrations only at antibiotic strips
 - BC: No flux on dish boundary
- **Results:**
 - Stable simulation with no losses.
 - Coherent Inhibition Concentration $c = 0.002 \mu\text{g}/\text{mm}^2$ similar to literature ($0.012 \mu\text{g}/\text{mm}^2$) [11]
- This confirms the validity of the approach for the model

Own Experimental Outcome



How Well Did Their
System Perform?

Biological Testing

Goal: Stress → System → Visual reporter

1) Response of different promoters towards different stressors

→ **stress reporters are functional**

→ **one stressor can activate several promoters**

2) Flow cytometry & confocal microscopy

→ **Output signal is detectable**

→ **Proof of concept for genetic circuit**

3) Fine tune dose-dependent responses

→ **Response is dose-dependent (threshold not shown)**

Synthetic Biology Success: completed core loop:

Design → Build → Test

Potential Issues:

- Stress responses: highly context dependent, can be non-specific
- freeze drying the cells might cause stress before testing

⇒ High-risk, high-reward biological target

Key Missing Elements:

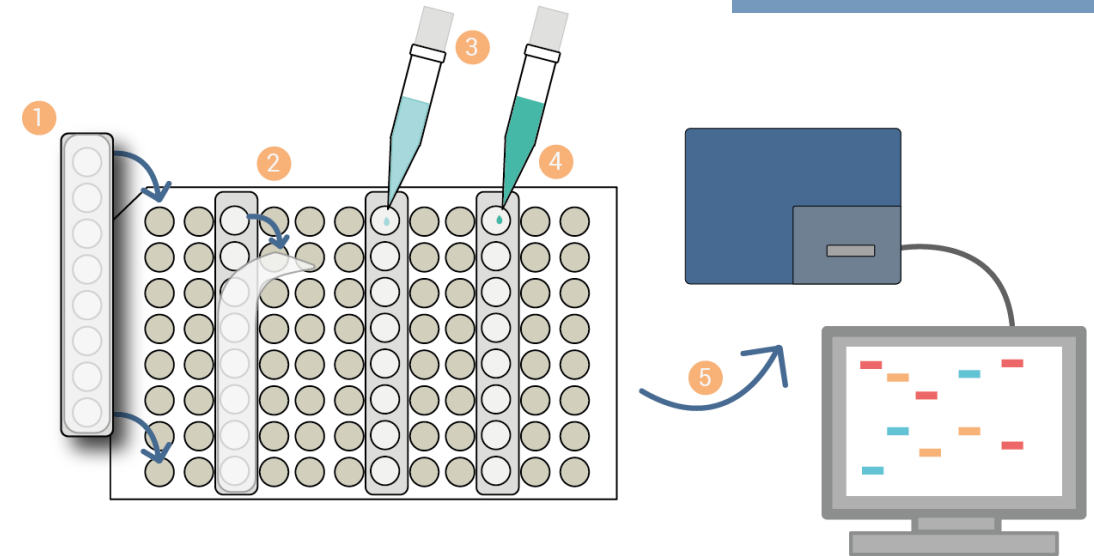
- **No Quantitative characterization:** dose-dependent curve; sensitivity threshold
- **Circuit robustness not determined:** reproducibility & stability over time
- **No real application test:** screening actual drug libraries

→ **Great prototype, but not (yet) a product**

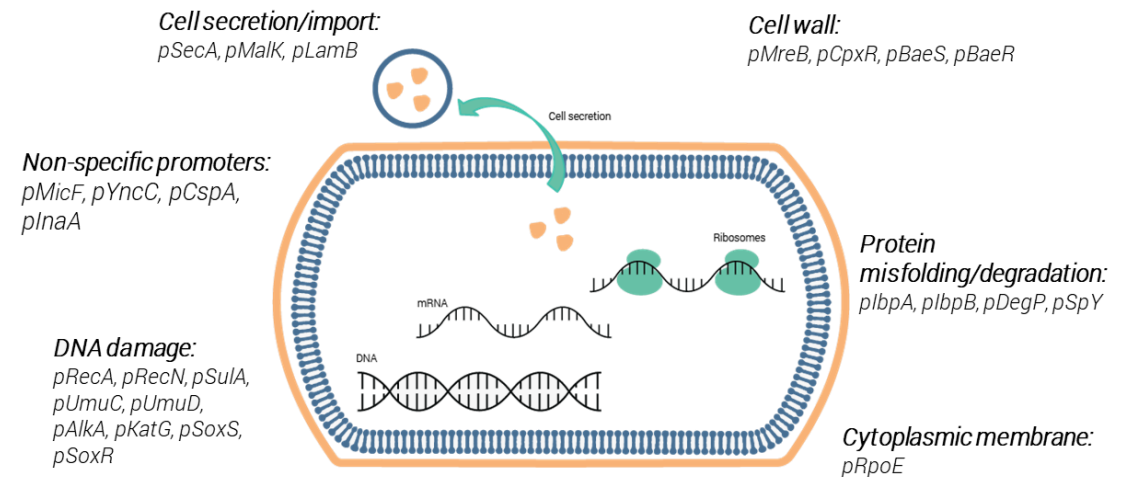
How Did They Intend to Design
the Finished Product?

- Compatible with standardized **96-well plates** → measurement in standard flow cytometers
- Customizable 8-well strips allow for various throughput needs
- Identification of **six stress categories** with two promoters each:
 - DNA damage,
 - Cell wall damage,
 - Protein misfolding,
 - Membrane damage,
 - Secretion/import stress,
 - Uncharacterized
- 3 unique promoters per well → **test one compound** towards all stress categories in **only 4 wells**

Product Design



Stress-activated promoters



Conceptual Innovation:

- Using stress as a screening proxy
→ Bridging synthetic biology & drug discovery

Strengths:

- Strong societal relevance
- Innovative screening platform
- Modular & extensive design
(high- & low-throughput)
- Platform potential
- Simple to use

Limitations:

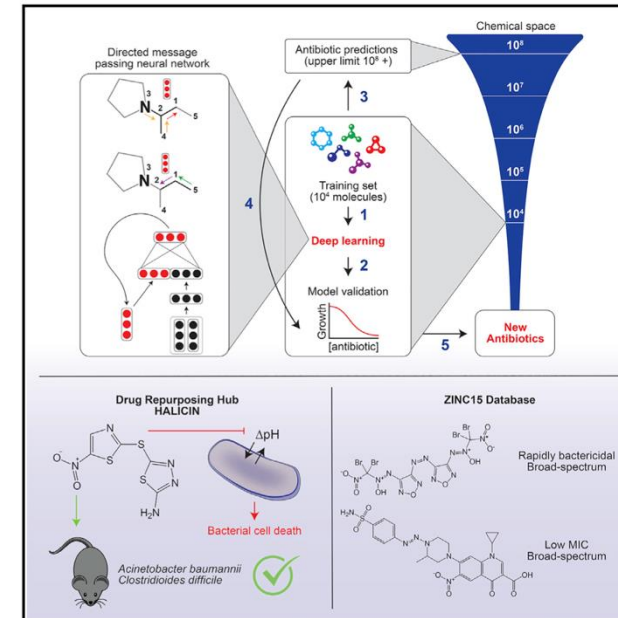
- Proof-of-concept stage
- Real-world application still far away
- Limited system characterization
- Biological complexity under-addressed

Conclusions

Cell

A Deep Learning Approach to Antibiotic Discovery

Graphical Abstract



Authors

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In Brief

A trained deep neural network predicts antibiotic activity in molecules that are structurally different from known antibiotics, among which Halicin exhibits efficacy against broad-spectrum bacterial infections in mice.

Only allows for testing of new antibiotics, but does not facilitate the finding of new candidate substances

Literature References

- [1] Murray C, Ikuta K, Sharara F et al., Global burden of bacterial antimicrobial resistance in 2019: a systematic analysis The Lancet, 2022; 399, 629-655
- [2] WHO 11.2023 - Antimicrobial resistance <https://www.who.int/news-room/fact-sheets/detail/antimicrobial-resistance>.
- [3] Muteeb, G.; Rehman, M.T.; Shahwan, M.; Aatif, M. Origin of Antibiotics and Antibiotic Resistance, and Their Impacts on Drug Development: A Narrative Review. Pharmaceuticals 2023
- [4] WHO 2025 - Global antibiotic resistance surveillance report 2025
- [5] McDonnell A et al. 2024 — Center for Global Development — Forecasting the Fallout from AMR: Economic Impacts of Antimicrobial Resistance in Humans
- [6] WHO 2024 - Antibacterial pipeline trends and recommendations to enhance research and development policy brief
- [7] WHO 2024 - Incentivizing the development of new antibacterial treatments. Progress Report by Global AMR R&D Hub & WHO
- [8] <https://2018.igem.org/Team:Leiden>
- [9] Kirby-Bauer Disk Diffusion Susceptibility Test Protocol (<https://asm.org/getattachment/2594ce26-bd44-47f6-8287-0657aa9185ad/kirby-bauer-disc-diffusion-susceptibility-test-protocol-pdf.pdf>)
- [10] Jonathan M. Stokes et al., A Deep Learning Approach to Antibiotic Discovery, Cell, Volume 180, Issue 4, 2020,
- [11] Alalam, Hanna. (2019). Re: What is the minimum inhibitory concentration of ampicillin for E. coli DH10B and DH5a?. Retrieved from: <https://www.researchgate.net/post/What-is-the-minimum-inhibitory-concentration-of-ampicillin-for-E-coli-DH10B-and-DH5a/5c379a27a7cbaf88b9282ec0/citation/download>.

Graphic References

[A] <https://2018.igem.org/Team:Leiden>

[B] WHO 2025 - Global antibiotic resistance surveillance report 2025

[C] Created with BioRender.com

[D] <https://www.sciencedirect.com/science/article/pii/S2221169117300436>

[E] Jonathan M. Stokes et al., A Deep Learning Approach to Antibiotic Discovery, Cell, Volume 180, Issue 4, 2020,

Thank You for Your Attention!
Are There Any Questions?